

# Rational interval quadrature as a corollary of Hilbert–Kamke theory

Literature consequence, not a claim of a new upper-bound theorem

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**Theorem 1.** *There is an absolute constant  $C$  such that for every  $r \geq 2$  and every integer  $K \geq Cr^3 2^r$ , rational nodes  $x_1, \dots, x_K \in (0, 1)$  exist with*

$$\frac{1}{K} \sum_{i=1}^K x_i^j = \frac{1}{j+1} \quad (0 \leq j \leq r).$$

*This follows from classical Hilbert–Kamke local-solubility results and the rational-node circle-method formulation of Cui–Xia–Xiang.*

**The classical local input.** Let  $V = (h^j)_{1 \leq j, h \leq r}$  and  $N \in \mathbb{Z}^r$ . Arkhipov’s Theorem 3 in [1], printed p. 34, supplies a positive uniform lower bound for the singular series if  $Vt = N$  is solvable in  $t \in \mathbb{Z}^r$  and

$$s > \min(r^2 2^{2r-1}, 3r^3 2^r - r).$$

In particular the congruences

$$\sum_{i=1}^s y_i^j \equiv N_j \pmod{p^a} \quad (1 \leq j \leq r)$$

are solvable for every prime  $p$  and every  $a \geq 1$ . The singular series is the product of the nonnegative local densities; absence of a solution modulo one prime power would make that local density zero. Compactness of  $\mathbb{Z}_p^s$  therefore gives a simultaneous solution in  $\mathbb{Z}_p^s$  for each  $p$ .

*Proof of Theorem 1.* Set  $c_j = K/(j+1)$  and  $B = |\det V|$ . Choose

$$P_0 = B(r+1)!.$$

For every positive multiple  $P$  of  $P_0$ , each number  $N_j = c_j P^j$  is an integer divisible by  $B$ . Thus  $V^{-1}N = (\text{adj } V)N / \det V$  is an integer vector. Arkhipov's local input, applied to this  $N$  with  $s = K$ , gives a solution  $y \in \mathbb{Z}_p^K$  for every  $p$ . Dividing by  $P$  gives a solution of  $\sum_i x_i^j = c_j$  in  $(P^{-1}\mathbb{Z}_p)^K$  for every  $p$ .

The real interval Chebyshev-quadrature theorem of Gilboa–Peled gives a real equal-weight rule of degree  $2r + 2$  for every sufficiently large  $K \geq C_0 r^2$ . Such a rule has at least  $r + 2$  distinct support points, by integrating the square of the polynomial vanishing on its support. There are therefore at least  $r$  distinct interior nodes. The first  $r$  moment Jacobian on these nodes is invertible. By the implicit function theorem, move any endpoint nodes inward and correct these  $r$  pivots. This gives a fixed real solution  $u \in (0, 1)^K$  of the first  $r$  moment conditions, with  $r$  distinct interior coordinates.

For all sufficiently large multiples  $P$  of  $P_0$ , this real solution lies in  $[P^{-1}, 1 - P^{-1}]^K$  and has positive separation quantity  $\Delta_u$  in [2, Theorem 5.2(v)]. That theorem's parts (iv) and (v) give positive uniform lower bounds for its singular series and singular integral. Part (i) then gives a positive number of rational solutions in  $((0, 1) \cap P^{-1}\mathbb{Z})^K$  for sufficiently large  $P$ . Its required variable threshold, polynomial in  $r$  times  $2^r$ , is covered by our choice  $K \geq Cr^3 2^r$ .  $\square$

**Attribution and sharper classical information.** The exponential upper bound is thus a direct consequence of earlier published theorems. The separate file `exponential_rational_designs.tex` supplies an independent quantitative local argument with the weaker threshold  $Cr^6 4^r$ . The interval-integral valuation argument giving an exponential lower bound is a separate statement; the unit-indicator polynomials used in that argument already occur in Hua, Arkhipov, and Mit'kin.

Mit'kin [4] sharpens the classical Hilbert–Kamke variable threshold to

$$H(r) = \sum_{0 \leq a \leq \lfloor \log_2 r \rfloor} 2^a (2^{\lfloor r/2^a \rfloor} - 1) = 2^r + O(2^{r/2}) \quad (r \geq 12).$$

The localized interval argument is now supplied in `mitkin_sharp_rational_upper.tex`: for every sufficiently large  $r$ , every  $K \geq H(r)$  admits a rational interval rule. In particular  $R_r \leq 2^r + O(2^{r/2})$ . The simpler argument above remains a direct application of Cui–Xia–Xiang with a larger threshold.

## References

- [1] G. I. Arkhipov, *On the Hilbert–Kamke problem*, Mathematics of the USSR-Izvestiya 24 (1985), no. 1, 1–47, doi:10.1070/IM1985v024n01ABEH001210.
- [2] Zhen Cui, Jiacheng Xia, and Ziqing Xiang, *Rational designs*, Advances in Mathematics 352 (2019), 541–571, doi:10.1016/j.aim.2019.06.012.
- [3] Shoni Gilboa and Ron Peled, *Chebyshev-type quadratures for doubling weights*, Constructive Approximation 45 (2017), 193–216, arXiv:1507.01505.
- [4] D. A. Mit’kin, *An estimate for the number of terms in the Hilbert–Kamke problem*, Mathematics of the USSR-Sbornik 57 (1987), no. 2, 561–590, doi:10.1070/SM1987v057n02ABEH003087.